Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
3	*(a)*	(i)	Use of the equation $\frac{M_2 d}{(M_1 + M_2)} = x$ even if wrong M_2 (1) $\frac{9.2 \times 10^{24} \times 5.3 \times 10^{10}}{(9.5 \times 10^{29} + \{9.2 \times 10^{24}\})}$ seen (1) ({ +9.2 × 10^{24} } can be omitted from the above equation) **or** 513 263 [m] seen etc.	1 1			2	2	
		(ii)	$v = \frac{2\pi r}{T}$ applied or $v = \omega r$ and $\omega = \frac{2\pi}{T}$ combined (1) Conversion of 130 days (130 × 24 × 3 600) (1) Answer = 0.287 [m s^{-1}] (**ecf** on day conversion) (1)	1	1 1		3	3	
		(iii)	Redshift is small **or** the redshift is proportional to velocity **or** small Δλ **or** shown using the Doppler equation (1) Small wavelength change is difficult to measure **or** shifted wavelength is too close to the original (1)		2		2		
	(b)	(i)	$F = \frac{GMm}{r^2}$ used (1) Answer = 2.08 × 10^{23} [N] (1)	1	1		2	2	
		(ii)	Method for obtaining stress i.e. $\frac{F}{A}$ (1) 1.35 × 10^9 [Pa] (1) Correct conclusion e.g. the steel bar would break **ecf** (1) Theory doesn't work / Newton's grav law used successfully for centuries / won't work for elliptical orbits / would melt (1) **Alternative:** Use of $F = \sigma A$ (1) F = 6.9 × 10^{22} [N] (1)			4	4	2	

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)	$\text{PE} = -\frac{GMm}{r}$ used N.B. **ecf** if PE = Fr used (1) Answer = [-]1.10 $\times$ 10^{34} [J] (1)	1	1		2	2	
		(ii)	Realising KE is involved (1) Correct variation with distance for KE or velocity **or** as PE increases KE decreases or converse (1)			2	2		
			Question 3 total	**5**	**6**	**6**	**17**	**11**	**0**